

REMOVE THE UNDIFFERENTIATED HEAVY LIFTING

If you run your code in containers on Amazon ECS and Amazon EKS, AWS is responsible for more of the container management, such as deploying containers across EC2 instances and managing the container cluster. However, when running ECS and EKS on EC2, you are still responsible for maintaining the underlying EC2 instances.

GO SERVERLESS

- No servers to provision or manage.
- Scales with usage.
- You never pay for idle resources.
- Availability and fault tolerance are built-in.

EXPLORE SERVERLESS CONTAINERS WITH AWS FARGATE

- Amazon EC2 mode
- AWS Fargate mode

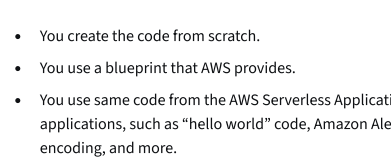
Select the cluster to run your task definition on and the number of copies of that task to run. To apply container overrides or target particular container instances, click **Advanced Options**.

It achieves this by allocating the right amount of compute, eliminating the need to choose and handle EC2 Instances and cluster capacity and scaling. Fargate supports both Amazon ECS and Amazon EKS architecture and provides workload isolation and improved security by design.

RUN YOUR CODE ON AWS LAMBDA

AWS Lambda requires zero administration from the user. You upload your source code and Lambda takes care of everything required to run and scale your code with high availability. There are no servers to manage, bringing you continuous scaling with subsecond metering and consistent performance.

There are three primary components of a Lambda function: the trigger, code, and configuration. The code is source code, that describes what the Lambda function should run. This code can be authored in three ways.



A trigger integrates your Lambda function with other AWS services, enabling you to run your Lambda function in response to certain API calls that occur in your AWS account. This makes you quicker to respond to events in your console without having to perform manual actions. All you need is the what, how, and when of a Lambda function have functional compute capacity that runs only when you need it to Amazon's CTO, Werner Vogels, says, "No server is easier to manage than no server." This quote summarizes the convenience you can have when running serverless solutions, like AWS Fargate and AWS Lambda.

AWS Lambda function handler

```
def handler_name(event, context): ... return some_value
```

The Lambda function *handler* name specified at the time you create a Lambda function is derived from the following: the name of the file in which the Lambda handler function is located the name of the Python handler function. A function handler can be any name; however, the default on the Lambda console is `lambda_function.lambda_handler`. This name reflects the function name as `lambda_handler`, and the file where the handler code is stored in `lambda_function.py`. If you choose a different name for your function handler on the Lambda console, you must update the name on the **Runtime settings** pane.

AWS Lambda lets you run code without provisioning or managing servers, and you pay only for what you use. You are charged for the number of times your code is triggered (requests) and for the time your code executes, rounded up to the nearest 1ms (duration). AWS rounds up duration to the nearest millisecond with no minimum execution time. With this pricing, it can be very cost effective to run functions whose execution time is very low, such as functions with durations under 100ms or low latency APIs. For more information, see [AWS News Blog](#).

This video used a small amount of sample code illustrating a pattern for lazily generating assets using AWS Lambda and Amazon S3. If you're looking to deploy a service to resize images to production, consider using the new release [Serverless Image Handler](#), which is a robust solution to handle image manipulation and can be deployed via an AWS CloudFormation template.

Resources

- [External Site: AWS: Serverless](#)
- [Coursera Course: Building Modern Python Applications on AWS](#)
- [External Site: AWS: AWS Serverless resources](#)
- [External Site: AWS: Building Applications with Serverless Architectures](#)
- [External Site: AWS: Best practices for organizing larger serverless applications](#)
- [External Site: AWS: Managing AWS Lambda functions](#)
- [External Site: AWS: 10 Things Serverless Architects Should Know](#)
- [External Site: AWS: AWS Alien Attack! A Serverless Adventure](#)

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